

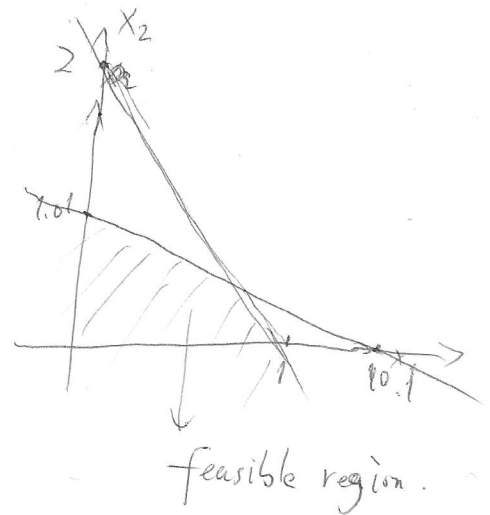
Barrier Methods (also known as interior-point methods) 是一种迭代方法

The core idea is to replace the inequality constraints with a smooth barrier function that penalizes points as they approach the boundary of the feasible region, while keeping iterates strictly inside the feasible set (the interior)

It enabled solutions of linear programming problems that were beyond the capacities of the simplex method. In contrast to the simplex method, it reaches a best solution by traversing the interior of the feasible region.

例子: $\max x_1 + x_2$
 $s.t. \quad 2x_1 + x_2 \leq 2 \quad \forall p \in [0, 0.1, 0.2 \dots 1.0]$

$2x_1 + x_2 \leq 2$
 $0.1x_1 + x_2 \leq 1.01$



问题 (P_b) min $f(x)$ $M[-g]$ 自变量记号
 $s.t. \quad x \in M[-g] = \{x \in \mathbb{R}^n \mid g_i(x) \geq 0, i \in I\}$

Lagrangian: $L(x, u) = f(x) - \sum_{i \in I} u_i g_i(x)$

We define $M_0 := \{x \in \mathbb{R}^n \mid g_i(x) > 0, i \in I\}$

and assume $M_0 = M^\circ$

定义: Barrier-function. Let $B: M_0 \rightarrow \mathbb{R}$ be continuous. B is called a barrier-function

if $\lim_{x \rightarrow \bar{x} \in \partial M} B(x) = +\infty$

M is a constraint set. $M[-g] =$ 可行域

Penalty method: 允许超出可行域
 Barrier method: 不允许在可行域里
 相同点: add ~~a penalty~~ a penalty like term to the objective function.